

Unit 6 – Inference for Categorical Data: Proportions
Notes 8 – Significance Test for a Difference in Proportions

Significance Test for a Difference between Two Proportions

Hypotheses

The way you will usually see it:

H_o :

H_a :

This is also a valid way:

H_o :

H_a :

Conditions

Normal

Both distributions must satisfy the large counts conditions:

Random

Both samples must have either:

- been randomly selected from the population
- used random assignment in an experiment

Independent

Both samples must satisfy the 10% condition:

$$n_1 \leq 0.1N_1$$

$$n_2 \leq 0.1N_2$$

*Only need to check for sampling

Pooled Proportion

_____ is the pooled sample proportion

- We use the pooled sample proportion to calculate the standard deviation because we always assume the null hypothesis of “no difference” is true, therefore, we assume the successes come from the same pool.
- We use this ONLY for significance tests
- We use this to check the large counts condition and to find the sample standard deviation

Calculating the P-Value

- Name the test: two proportion z test
- Calculate the test statistic:

OR

- Find your p-value using normalcdf

Example: Two Proportion Z Test

Is there a significant difference in the smoking habits of the younger and older generation? In a random sample of 600 young adults (aged 18 to 22), it was found that 180 of them had smoked in the past week. In a separate random sample of 500 older adults (aged 50 and above), it was found that 185 of them had smoked in the past week. Conduct a significance test at the 5% level to determine if there is evidence to suggest that the proportion of smokers differs between young adults and old adults.

a) What type of error could we have made in the example above?

b) Find a 95% confidence interval using your calculator. Interpret it and explain how it confirms your results from above.

Two Proportion Z Test On the TI-84

Stat → Tests → 6: 2-Prop Z Test

Asks for:

x_1, n_1, x_2, n_2

$p_1 : \neq p_2 > p_2 < p_2$

Returns:

z, p (p-value), $\hat{p}_1, \hat{p}_2, \hat{p}$ (pooled $\hat{p}_C$)

